

Explainable NLP

Explainability vs Interpretability

- **Interpretability:** How well can a human understand the model?
 - **Explainability:** How well can the model explain its decisions?
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Why explain?

- **Transparency:** Understand how the model works
 - **Trust:** Trust the model's decisions
 - **Bias:** Detect and correct biases
 - **Regulation:** Comply with regulations, e.g. GDPR
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Performance vs Explainability

- Performance (accuracy) is often at odds with explainability
- Black-box models (e.g. deep learning) are often more accurate
- Traditional methods (e.g. decision trees) are more interpretable
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Discussion: What is meant by “black-box”?

Fidelity, plausibility and usability

- **Fidelity:** How well does the explanation reflect the model?
 - **Plausibility:** How well does the explanation make sense?
 - **Usability:** How well can the explanation be used?
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Post-hoc explanations

- **Post-hoc:** Explain after the model has made a decision
 - Explanation does not follow the path the model took
 - As such, post-hoc explanations are not always accurate
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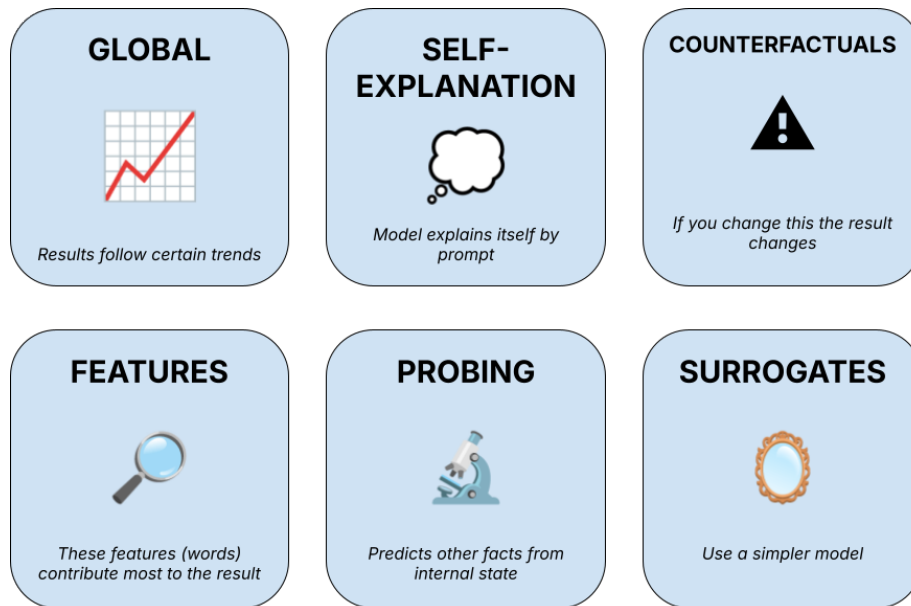


Figure 1: Explanation Types

Local vs Global Explanations

- **Local**: Explain a single prediction
 - **Global**: Explain the model as a whole
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Global Explanations

- Provide an overall understanding of the model's behavior
 - For example, feature importance across the entire dataset
 - Global explanations can explain biases in the model
 - **Explanations over multiple data points**
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Self-explanation

- Simply prompt the model to explain its decision
 - Explanations tend to be highly plausible
 - But may not reflect the model's actual decision
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Self-explanation example

Prompt: “What is the sentiment of the following review? Explain your reasoning. ‘The movie was fantastic!’ ”

Model output: “The sentiment of the review is positive. The word ‘fantastic’ indicates a strong positive sentiment towards the movie.”

Example-based

- Example-based explanations illustrate how a model’s output changes with different examples
 - **Counterfactual example:** By perturbations or with another generative model, find examples that affect the output
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Counterfactual example

- “The movie was **fantastic** and the acting was **superb!**” (positive)
 - “The movie was **fantastic** and the acting was **terrible!**” (negative)
 - “The movie was **terrible** and the acting was **superb!**” (negative)
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Counterfactuals

- Counterfactuals can be useful for debugging and understanding model behavior
 - However, they may not always be realistic or plausible
 - Example:
 - “The movie was fantastic **but** the acting was superb!” (negative)
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Intrinsic explanations

- **Intrinsic:** Explain while the model is making a decision
 - Explanation follows the path the model took
 - Explanations may make less sense to humans
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Feature attribution

- Feature attributions methods measure the relevance of input features
 - **Perturbation:** Change the input and measure the change in output
 - **Gradient-based:** Use gradients to measure feature importance

- **Decomposition:** Decompose the model to measure feature importance

Attention features

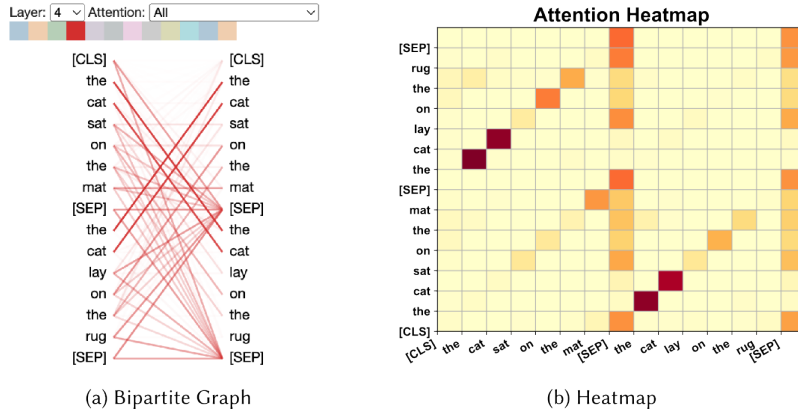


Figure 2: Attention Heatmap

Image source: Explainability for Large Language Models: A Survey, Zhou et al., 2024

Probing

- Train a shallow classifier on top of the model
- For example:
 - Train a part-of-speech on each layer of an LLM
 - Deduces when model has learned syntax

Conceptual explanations

- Map the inputs to a set of concepts
- For example:
 - For sentiment analysis, map features to adjectives
 - Use adjective sentiment lexicons to explain results

Mechanistic Interpretability

- Explain the model in terms of **circuits**
 - A circuit is a subgraph of the neural network
 - Example is an **induction head**:
 - One attention head copies information from a previous token
 - A second head copies information from a previous token
 - Circuits are difficult to find as features are spread across the network in **superposition**
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Surrogate models

- Train a simpler model to approximate the behavior of a complex model
 - For example, train a decision tree to approximate a deep learning model
 - Surrogates have **very low fidelity**
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Discussion

- Feature attribution
- Probing
- Example-based
- Conceptual explanations

Which of these methods are local and global?

Future Directions

- Do explanations make LLMs safer?
- Could explanations be interactive?
- How could neurosymbolic approaches help?